CS 4644/7643: Deep Learning
Fall 2026
Problem Set 3

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Discussions: https://piazza.com/class/ms43b2tjxa63lq#

Due: October 28, 2026, 11:59pm

Instructions

1. We will be using Gradescope to collect your assignments. Please read the following instructions for submitting to Gradescope carefully!

   - For the **HW3 Theory** component on Gradescope, upload one single PDF containing the answers to all the theory questions. **However, the solution to each problem/subproblem must be on a separate page. When submitting to Gradescope, please make sure to mark the page(s) corresponding to each problem/subproblem.** Questions with misassigned pages will be penalized by up to 1 point.
   - For the **HW3 Code** on Gradescope, please use the `collect_submission.py` script provided and upload **hw3_code_submission.zip**
   - For the **HW3 Notebook** component on Gradescope, please use the `collect_submission.py` script provided and upload the resulting **hw3_notebook_submission.pdf** on Gradescope.
   - Note: This is a large class and Gradescope's assignment segmentation features are essential. Failure to follow these instructions may result in parts of your assignment being penalized. We will not entertain regrading requests for failure to follow instructions.

2. LaTeX'd solutions are strongly encouraged (solution template available on Canvas—HW3 Theory), but scanned handwritten copies are acceptable. Hard copies are **not** accepted.

3. We generally encourage you to collaborate with other students.

   You may talk to a friend, discuss the questions and potential directions for solving them. However, you need to write your own solutions and code separately, and *not* as a group activity. Please list the students you collaborated with.

1 Collaborators [0.5 points]

Please list your collaborators and assign this list to the corresponding section of the outline on Gradescope. If you don't have any collaborators, please write 'None' and assign it to the corresponding section of the Gradescope submission regardless. Not assigning this question on Gradescope or leaving it blank will be marked as zero.

2 Logic and XOR

1. **[1 point]** Implement AND and OR for pairs of binary inputs using a single linear threshold neuron with weights $\mathbf{w} \in \mathbb{R}^2$, bias $b \in \mathbb{R}$, and $\mathbf{x} \in \{0,1\}^2$:

$$f(\mathbf{x}) = \begin{cases} 1 & \text{if } \mathbf{w}^T\mathbf{x} + b \geq 0 \\ 0 & \text{if } \mathbf{w}^T\mathbf{x} + b < 0 \end{cases} \tag{1}$$

   That is, find $\mathbf{w}_{\texttt{AND}}$ and $b_{\texttt{AND}}$ such that

x_1	x_2	$f_{\texttt{AND}}(\mathbf{x})$
0	0	0
0	1	0
1	0	0
1	1	1

   Also find $\mathbf{w}_{\texttt{OR}}$ and $b_{\texttt{OR}}$ such that

x_1	x_2	$f_{\texttt{OR}}(\mathbf{x})$
0	0	0
0	1	1
1	0	1
1	1	1

   You do not need to show your work for this question. You can just provide the specific values for $\mathbf{w}$ and b.

2. **[1 point]** Consider the XOR function

x_1	x_2	$f_{\texttt{XOR}}(x)$
0	0	0
0	1	1
1	0	1
1	1	0

   Prove that XOR can NOT be represented using a linear model with the same form as (1). Graphical/visual proofs will not receive full credit.

   [*Hint:* To see why, plot the examples from above in a plane and think about drawing a linear boundary that separates them.]

3 Recurrent Neural Networks and Transformers

3. **[LSTM for parity function: 4 points]** Let us recall different gates in an LSTM Network. First gate is the "forget gate layer":

$$f_t = \sigma(W_f \cdot \begin{bmatrix} h_{t-1} \\ x_t \end{bmatrix} + b_f) \tag{2}$$

where f_t is the output of forget gate, W_f is the weight matrix, h_{t-1} is the hidden state of step *t-1*, x_t is the current input and b_f is the bias value.
Next we have "input gate layer":

$$i_t = \sigma(W_i \cdot \begin{bmatrix} h_{t-1} \\ x_t \end{bmatrix} + b_i) \tag{3}$$

$$\tilde{C}_t = \tanh(W_C \cdot \begin{bmatrix} h_{t-1} \\ x_t \end{bmatrix} + b_C) \tag{4}$$

where i_t decides which values we will update and $\tilde{C}_t$ are the new candidate values that could be added to the cell state. Next we have new cell state candidate values:

$$C_t = C_{t-1} \times f_t + \tilde{C}_t \times i_t \tag{5}$$

Finally, we have the output and hidden state

$$o_t = \sigma(W_o \cdot \begin{bmatrix} h_{t-1} \\ x_t \end{bmatrix} + b_o) \tag{6}$$

$$h_t = o_t \times \tanh(C_t) \tag{7}$$

Design an LSTM Network for the bit parity problem, which returns 0 if there is an even number of 1s in an input bit string, and 1 otherwise. Specifically,

(a) provide values for W_f, b_f, W_i, b_i, W_C, b_C, W_o and b_o such that the cell state C_t stores the parity of an input bit string, assuming x, h, f, i, o, b, and C are all scalars.

(b) briefly summarize how your LSTM implements the logic to determine bit parity

(c) explain how you obtained your values

Please mention any assumptions you make. For this problem, you can assume below for Sigmoid and tanh function:

$$\sigma(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

$$\tanh(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & x = 0 \\ -1, & \text{if } x < 0 \end{cases} \tag{9}$$

Hint: Recall that XOR of x and y can be represented as $(x \wedge \bar{y}) \vee (\bar{x} \wedge y)$ which you implemented above. Think about how you can leverage this information for equation (5).

4. **[When to Stop in Beam Search: 2 points]** Beam Search is a widely-used technique for decoding the most likely sequence from sequence models. But it is difficult to decide when to stop beam search to obtain optimality because hypotheses can finish in different steps. In this question, we will develop a formal understanding of the stopping criteria in beam search.

   Suppose we are generating a sequence $\mathbf{y} = (y_1, y_2, \ldots, y_T)$ conditioned on input $\mathbf{x}$ using beam search with beam size b, which is the number of sequences we are exploring in parallel. We will sample next tokens in the sequence using the distribution defined by $p(y_t|\mathbf{x}, y_{<i})$, which can be thought of as an autoregressive sequence model.

   - A sequence $\mathbf{y}$ is **complete** if it ends with `</s>`:

$$\textsc{completed}(\mathbf{y}) = \text{true} \iff y_{|\mathbf{y}|} = \texttt{</s>}$$

- The **score** $s(\mathbf{y})$ of a sequence $\mathbf{y}$ is the sum of the token log probabilities:

$$s(\mathbf{y}) = \sum_{i=1}^{|\mathbf{y}|} \log p(y_i \mid \mathbf{x}, y_{<i})$$

- Let B_t be the set of the top-b active beams at step t, where each element is $\langle \mathbf{y}, s(\mathbf{y}) \rangle$.

$$B_t = \text{top}^b \Big\{ \langle \mathbf{y} \circ y_t,\ s(\mathbf{y}) + \log p(y_t \mid \mathbf{x}, \mathbf{y}) \rangle \ : \langle \mathbf{y}, s(\mathbf{y}) \rangle \in B_{t-1},\ y_t \in \mathcal{V} \Big\}$$

  where $\mathcal{V}$ is the vocabulary and top^b selects the b highest-scoring candidates. Assume we start sampling from an empty sequence such that $B_0 = \{\langle \{\}, 0 \rangle\}$.

- $s^{(t)}_{\text{best}}$ = score of the best *completed* sequence found up to step t:

$$s^{(t)}_{\text{best}} = \max \left\{ s(\mathbf{y}) \mid \langle \mathbf{y}, s(\mathbf{y}) \rangle \in \bigcup_{j \le t} B_j,\ \textsc{completed}(\mathbf{y}) = \text{true} \right\}$$

  Suppose at step t, $s^{(t)}_{\text{best}}$ is defined (at least one complete sequence exists) and that the current highest scoring beam in B_t (*i.e.* $\mathbf{y}^1$) scores worse than or equal to $s^{(t)}_{\text{best}}$ (*i.e.* $s^1 \le s^{(t)}_{\text{best}}$). **For this question, explain why no future completed sequence $\mathbf{y}^*$ (of length $>= t+1$) will have a score greater than $s^{(t)}_{\textbf{best}}$.**

5. **[Exploding Gradients: 2 points]** Learning long-term dependencies in recurrent networks suffers from a particular numerical challenge – gradients propagated over many time-steps tend to either 'vanish' (converge to 0) or 'explode' (diverge to infinity). To study this problem in a simple setting, consider the following recurrence relation without any nonlinear activation function or input x:

$$h_t = W h_{t-1} \tag{10}$$

where W is a weight sharing matrix for recurrent relation at any time t. Let $\lambda_1, ..., \lambda_n$ be the n distinct eigenvalues of the weight matrix $W \in \mathbb{C}^{n \times n}$. Its spectral radius $\rho(W)$ is defined as:

$$\rho(W) = \max\{|\lambda_1|, ..., |\lambda_n|\} \tag{11}$$

Assuming the initial hidden state is h_0, write h_t in terms of h_0 and W and explain the role of the eigenvalues of W in determining the 'vanishing' or 'exploding' property as $t \gg 0$ (t grows very large).

4 Paper Review

For this homework's paper review section, we will read **FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness**. The paper can be viewed here.
The evaluation rubric for this section is as follows:

6. **[2 points]** Briefly summarize the key contributions, strengths and weaknesses of this paper.

7. **[2 points]** What is your personal takeaway from this paper? This could be expressed either in terms of relating the approaches adopted in this paper to your traditional understanding of explainability techniques, or potential future directions of research in the area which the authors haven't addressed, or anything else that struck you as being noteworthy.

Guidelines: Please restrict your reviews to at least one paragraph per question, but no more than 350 words (total length for answers to both of the above questions). Please write separate answers for each question to assist in Gradescope grading.